Polylogarithmic Approximation for Covering and Connecting Multi-Interface Networks

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Abstract

We consider the problem of covering/connecting a multi-interface network with a minimum cost assignment of interfaces to vertices. In the covering problem we demand that every edge is covered by at least one common interface assigned to its endpoints. In the connection problem we are given a set of terminals $U \subseteq V$ and we demand that the subgraph of activated edges spans all of them. We give an $O(\log m)$ approximation algorithm for the min-max cost criterion. The algorithm is based on rounding the optimal solution of a natural, set-cover-style linear programming relaxation of the problem. We also show how to use this LP to obtain a k-approximation algorithm, where k is the number of interfaces. Importantly, both of the aforementioned procedures work even when the cost of activating an interface depends on the interface as well as the vertex to which it is assigned. This improves the previously best known $O(\ln \Delta + b \cdot \min\{\ln \Delta, c_{\max}\})$-approximation, where b is a certain parameter of the input instance which can be as large as $\Omega(n)$ and $c_{\max}$ is the maximum cost of activating an interface and $O(k)$-approximation for the uniform cost case [Algorithmica 2012]. For the connection problem we devise an $O\big(\log^2 m\big)$-approximation algorithm, which is the first non-trivial approximation algorithm for this problem. The algorithm is based on a similar LP relaxation with additional cut constraints to ensure connectivity of the terminals.

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1 Introduction

Designing Internet of Things (IoT) networks often demands establishing connection between large number of devices, which can be achieved by choosing between large variety of interfaces available at each device. Moreover, the devices present in the network may vary according to the interfaces they provide for the usage. For example some device may be able to connect by Bluetooth and Wi-Fi, while the others may provide Zigbee and Z-Wave interfaces. Since IoT devices are typically resource-constrained, it may be desirable to try to establish every relevant connection, while minimizing the amount of resources required to do so.

We model the above problem as a graph problem. Every device in the network is modeled by a vertex with a set of labels representing its available interfaces. Two devices can communicate if they share a common available interface and are sufficiently close to each other, and the edges of the graph represent the possible connections. A connection is established between two devices when they share a common active interface. We want to find an assignment of active interfaces so that we ensure either coverage or connectivity of the network. In the coverage problem, we wish to establish every possible connection in the network (see Figure 1 for an example) while in the connectivity problem, we are additionally given a subset of vertices called terminals, an we require that terminals belong to the same connected component in the subgraph induced by active interfaces (see Figure 2 for an example). In both problems, we assume that activating an interface induces a cost that depends on the type of interface and on the vertex that activates it. The goal is to

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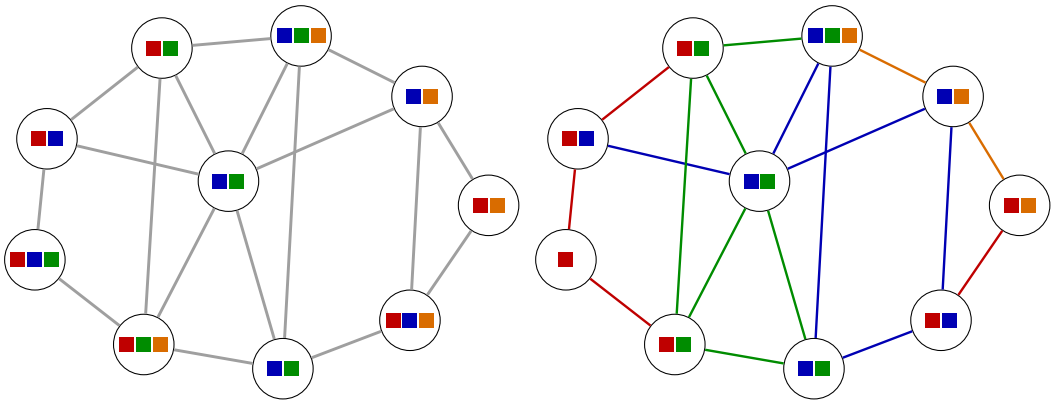
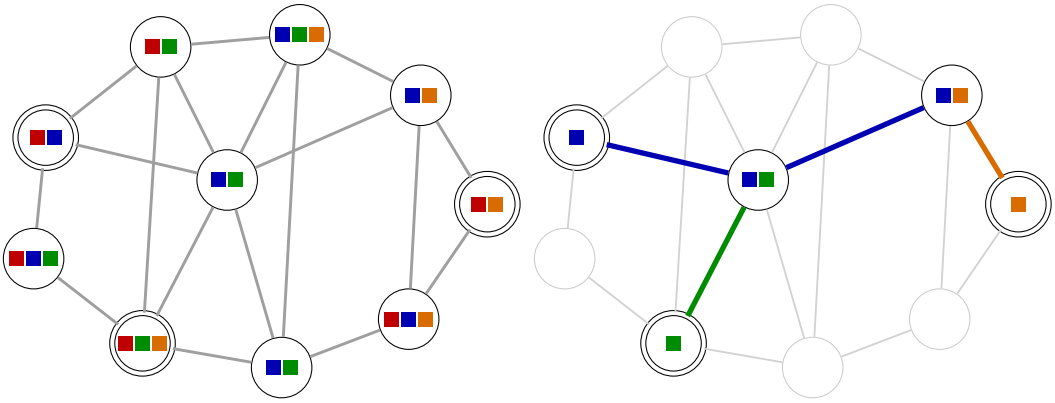


Available labels

Example covering assignment

Figure 1 An example instance of a multi-interface network with available interfaces at each vertex and a feasible covering assignment.



Available labels (double contour = terminal)

Example connecting assignment

Figure 2 An example instance of a multi-interface network with available interfaces at each vertex and a feasible connecting assignment.

minimize either the min-sum cost (the sum of costs over all vertices) or the min-max cost (the maximum cost in a vertex).

1.1 Related Work

In the following, we denote k the number of different interfaces across the network, n the number of vertices, Δ the maximum degree of the graph, $c_{\max}$ the maximum cost and $c_{\min}$ the minimum cost of an interface. In the *homogeneous* setting, the costs depend on the interface only while in the *heterogeneous* case they depend on the interface and on the vertex.

Coverage Problem. The algorithmic study of problems in multi-interface networks started with the *coverage* problem, introduced in Caporuscio et al. [6]. The coverage problem has been studied only in the homogeneous setting, both for min-sum [14] and min-max [9] objective functions.

The min-sum variant is APX-hard for any constant $k \geq 3$ and unit costs, and it is hard to approximate within a $O(\ln k)$ factor even with unit costs, but it has a $\min\{k-1, \sqrt{n}(1+\ln n)\}$-

approximation [14].

The min-max variant is NP-Hard for any constant $\Delta \geq 5$, $k \geq 16$ and for unit costs, and it is not approximable within a $\eta \ln \Delta$ factor for some constant η even on trees and with unit costs. On the positive side, it admits a $(\ln \Delta + 1 + b \cdot \min\{c_{\max}, \ln \Delta + 1\})$-approximation, where b is a parameter that depends on the structural properties of the input graph and is bounded by many other graph structutal parameters such as treewidth, arboricity or maximum degree. It also has a $(1 + (k-2)\frac{c_{\max}}{2c_{\min}})$-approximation and a $(\frac{\Delta}{2}\frac{c_{\max}}{c_{\min}})$-approximation [9].

Connectivity Problem. The *connectivity* problem was introduced by Kosowski et al. [16]. It was first studied in the homogeneous setting and in the special case where all vertices are terminals, both for the min-sum [16] and the min-max objective functions [9].

The min-sum variant with terminals only and homogeneous costs is APX-hard for $k \geq 2$, $\Delta \geq 4$ and unit costs already, but has a 2-approximation [16]. This result was improved by Athanassopoulos et al. [4] to a randomized $(\frac{3}{2} + \epsilon)$-approximation for any constant ϵ. Moreover, this approximation algorithm applies to the more general problem where for each interface, we are additionally given the set of edges that it can cover.

The min-max variant where costs are homogeneous and all vertices are terminals is NP-Hard for any constant $\Delta \geq 3$, $k \geq 10$ and for unit costs. Similarly to the coverage problem, it is not approximable within a $\eta \ln n$ factor even for unit costs, but has simple $(1 + (k-2)\frac{c_{\max}}{2c_{\min}})$- and $\frac{\Delta}{2}\frac{c_{\max}}{c_{\min}}$-approximations [9].

Athanassopoulos et al. [4] expand the study of the min-sum connectivity problem to *group connectivity*, a more general problem where the solution graph should connect together groups of vertices that might not span the whole set of vertices. They provide a randomized 4-approximation for it. They also initiate the study of this problem in the heterogeneous setting, where costs can also depend on the vertex. They show that the heterogeneous connectivity problem is hard to approximate within $o(\ln n)$, but the group connectivity heterogeneous case admits a $O(\ln n)$-approximation.

Other Multi-interface Problems. There has been some work on the study of the coverage problem from the perspective of parameterized complexity, possibly with an addtional constraint on the maximum number of active interfaces in a vertex [2, 1, 5], yielding efficient algorithms in some graph classes. Some papers also study other problems in multi-interface networks, like cheapest path [16], matchings [15], or flow problems [10].

1.2 Our contribution and techniques

In this paper we give a first $O(\log m)$-approximation algorithm for the min-max connection problem with non-uniform costs. This improves the previously known result of [9], which gave an $O(\ln \Delta + b \cdot \min\{\ln \Delta, c_{\max}\})$-approximation for the min-max covering version, where b is the parameter related to the existence of a b-owner function [11, 8]. This als improves the $O(\sqrt{n} \cdot \log n)$-approximation for the min-sum covering problem by [14]. Our approach is based on a set-cover-style LP relaxation together with a randomized rounding. We also show how to use the same LP relaxation to obtain a k-approximation algorithm for the min-sum and min-max covering problems, where k is the number of interfaces. All these guarantees hold in the non-uniform setting where activation costs depend on both the interface and the vertex it is assigned to.

2 Preliminaries

Let $G = (V, E)$ be a connected simple graph. Let $[k]$ be the set of different types of interfaces across the network and let $\lambda\colon V \to [k]$ be a function that assigns to each vertex $v \in V$ its set of available interfaces such that for each edge $uv \in E$, $\lambda(u) \cap \lambda(v) \neq \emptyset$. An assignment of active interfaces is a function $\lambda_A\colon V \to [k]$ such that $\lambda_A(v) \subseteq \lambda(v)$ for all $v \in V$. We say that an edge $uv \in E$ is *covered* if $\lambda_A(u) \cap \lambda_A(v) \neq \emptyset$ and we denote by G_A the subgraph of G induced by covered edges.

In the *coverage problem*, we wish to find an assignment of active interfaces λ_A such that $G_A = G$. We call such an assignment a *covering assignment*.

In the *connectivity problem*, we are additionally given a set of *terminals* $U \subseteq V$ and we wish to find an assignment of active interfaces λ_A such that all vertices in U lie in the same connected component of G_A. Vertices in $V \setminus U$ may be used as Steiner vertices. We call such an assignment a *connecting assignment*. The special case $U = V$ corresponds to requiring that G_A is a connected subgraph of G spanning V.

We are also given a cost function $c\colon [k] \times V \to \mathbb{N}_{\geq 0}$, which denotes the cost of activating the interface $i \in [k]$ in the vertex $v \in V$. The sum cost of an assignment λ_A is defined as:

$$\texttt{COST}_S = \sum_{v \in V} \sum_{i \in \lambda_A(v)} c(i, v)$$

and the max cost of and assignment λ_A is defined as:

$$\texttt{COST}_M = \max_{v \in V} \sum_{i \in \lambda_A(v)} c(i, v)\,.$$

The goal of the min-sum (resp. min-max) assignment problem is to find an assignment λ_A that minimizes $\texttt{COST}_S$ (resp. $\texttt{COST}_M$). The considered problems are formalized below:

Coverage Problem

Input: A connected simple graph $G = (V, E)$, available interfaces $\lambda\colon V \to [k]$, a cost function $c\colon [k] \times V \to \mathbb{N}_{\geq 0}$.
Task: Find a min-sum (resp. min-max) covering assignment λ_A minimizing $\texttt{COST}_S$ (resp. $\texttt{COST}_M$).

Connectivity Problem

Input: A connected simple graph $G = (V, E)$, available interfaces $\lambda\colon V \to [k]$, a cost function $c\colon [k] \times V \to \mathbb{N}_{\geq 0}$, set of terminals $U \subseteq V$.
Task: Find a min-sum (resp. min-max) connecting assignment λ_A minimizing $\texttt{COST}_S$ (resp. $\texttt{COST}_M$).

Additionally, we need the following probability inequality known as the Chernoff bound [7]:

▶ **Theorem 1** (Chernoff Bound). *Let $X_1, \ldots, X_n$ be independent random variables taking values in $\{0, 1\}$, $w_1, \ldots, w_n \in [0, 1]$ be weights, and let $X = \sum_{i=1}^{n} w_i X_i$. Then for every $\delta > 0$:*

$$\Pr[X \geq (1 + \delta) \cdot \mathbb{E}[X]] \leq \left(\frac{e^{\delta}}{(1 + \delta)^{1+\delta}} \right)^{\mathbb{E}[X]}$$

and for every $0 < \delta < 1$*:*

$$\Pr[X \leq (1-\delta)\cdot \mathbb{E}[X]] \leq \left(\frac{e^{-\delta}}{(1-\delta)^{1-\delta}}\right)^{\mathbb{E}[X]}.$$

3 LP based approximation algorithms

Below we show an $O(\log m)$ approximation algorithms for all four variants and $O(k)$-approximation for the covering problems. The idea of our ILP formulation and the algorithm is inspired by the rounding procedure known for set cover [17] and related problems [3]

3.1 Rounding the costs

Due to technical reasons, we show how to produce a polynomial number of instances, with cost function bounded by a polynomial in n. By testing consecutive powers of 2, we find the smallest number b, such that the costliest activated interface in the optimal solution has cost at most 2^b. For each vertex $v \in V$, we discard all of the interfaces from $\lambda(v)$ with cost larger than 2^b and we normalize the costs of remaining interfaces by dividing them by 2^b. Note that the number of guesses is polynomial. We also can assume that $\texttt{OPT} \geq 1/2$, since the cost of any solution is at least the cost of activating one interface, which is at least $1/2$ by our rounding.

3.2 ILP formulation of covering multi-interface networks

We start with the following ILP formulation of the min-sum covering assignment problem. Let $x_{i,v}$ be a binary variable that is 1 if and only if the interface $i \in \lambda(v)$ is assigned to vertex $v \in V$. We consider two types of vertices. For a given vertex $v \in V$, if $\sum_{i\in\lambda(v)} c(i,v) \leq 1$, then we can safely assume that all interfaces in $\lambda(v)$ are activated at v, since the cost of activating all of them is at most 1 and thus at most $2 \cdot \texttt{OPT}$. We call such vertices *cheap*. For the remaining vertices, we call them *expensive*. Note that for every expensive vertex v, we can assume that $\sum_{i\in\lambda(v)} x_{i,v} \cdot c(i,v) \geq 1$, since this will always result in a solution of cost at most $2 \cdot \texttt{OPT}$. Thus, we can write the following ILP formulation of the min-max covering assignment problem:

$$\begin{aligned}
\min \quad & M \\
\text{s.t.} \quad & \sum_{i\in\lambda(v)} x_{i,v}\cdot c(i,v) \leq M \quad \forall v \in V \\
& \sum_{i\in\lambda(u)\cap\lambda(v)} \min\{x_{i,u}, x_{i,v}\} \geq 1 \quad \forall uv \in E \\
& \begin{cases} x_{i,v} = 1 \quad \forall i \in \lambda(v) & \text{if } \sum_{i\in\lambda(v)} c(i,v)\cdot x_{i,v} < 1 \\ \sum_{i\in\lambda(v)} c(i,v)\cdot x_{i,v} \geq 1 & \text{otherwise} \end{cases} \quad \forall v \in V \\
& x_{i,v} \in \{0,1\} \quad \forall v \in V, i \in \lambda(v)
\end{aligned}$$

Note that the above formulation is not linear due to the presence of the min function. However, by adding an additional variable $y_{i,uv}$ we can linearize the constraints of type $\sum_{i\in\lambda(u)\cap\lambda(v)} \min\{x_{i,u}, x_{i,v}\} \geq 1$ by replacing them with $\sum_{i\in\lambda(u)\cap\lambda(v)} y_{i,uv} \geq 1$ together with the constraints $y_{i,uv} \leq x_{i,u}$ and $y_{i,uv} \leq x_{i,v}$ for every $i \in \lambda(u) \cap \lambda(v)$ and $uv \in E$. Thus, we can solve the LP relaxation of this formulations in polynomial time.

3.3 Randomized $O(\log m)$-approximation algorithm for the covering problem

Algorithm 1 A relaxation-based $O(\log m)$-approximation algorithm.

1. Solve the LP relaxation and obtain a fractional solution $\{x_{i,v}\}_{i\in\lambda(v),v\in V}$.
2. Pick independent uniformly random thresholds $t_i \in (0,1)$, for each $i \in [k]$.
3. Set $\lambda_A(v) = \{i \in \lambda(v) \colon 3\ln m \cdot x_{i,v} \geq t_i\}$, for every $v \in V$.
4. **return** $\{\lambda_A(v)\}_{v\in V}$.

▶ **Theorem 2.** *The above algorithm is an $O(\log m)$-approximation algorithm for the min-max covering assignment problems which succeeds with high probability.*

Proof. We have the following lemmas:

▶ **Lemma 3.** *The algorithm returns a solution of cost at most $9\ln m \cdot \mathtt{OPT}$ with high probability.*

Proof. Fix a vertex $v \in V$. We want to show that with high probability, the cost of activating interfaces at v is at most $9\ln m$ times the cost of activating interfaces at v in the fractional solution. If v is cheap, then the cost of activating all interfaces at v is at most 1, so the claim holds. Assume that v is expensive. By $X_{v,i}$ denote the random variable that is 1 if $3\ln m \cdot x_{i,v} \geq t_i$ and 0 otherwise. Then we can write $c(\lambda_A(v)) = \sum_{i\in\lambda(v)} X_{v,i} \cdot c(i,v)$. We have: $\mathbb{E}[X_{v,i}] = \Pr[i \in \lambda_A(v)] = 3\ln m \cdot x_{i,v}$. Thus, $\mathbb{E}[c(\lambda_A(v))] = 3\ln m \cdot \sum_{i\in\lambda(v)} x_{i,v} \cdot c(i,v)$. By applying the Chernoff bound with $\delta = 2$ we get that

$$\Pr[c(\lambda_A(v)) \geq 3 \cdot \mathbb{E}[c(\lambda_A(v))]] \leq \left(\frac{e^2}{3^3}\right)^{\mathbb{E}[c(\lambda_A(v))]} \leq \exp\{-\mathbb{E}[c(\lambda_A(v))]\} \leq \frac{1}{m^3}$$

where the last inequality is by using the fact that, $\sum_{i\in\lambda(v)} x_{i,v} \cdot c(i,v) \geq 1$. By applying the union bound over all n vertices we get that with probability at least $1 - \frac{n}{m^3} \geq 1 - \frac{1}{m}$ (assuming $m \geq 2$) for every $v \in V$, $c(\lambda_A(v)) \leq 9\ln m \cdot \sum_{i\in\lambda(v)} x_{i,v} \cdot c(i,v)$. Thus, with high probability we have $\mathtt{COST}(\lambda_A) \leq 9\ln m \cdot \max_{v\in V} \sum_{i\in\lambda(v)} x_{i,v} \cdot c(i,v) = 9\ln m \cdot \mathtt{OPT}$. ◀

▶ **Lemma 4.** *With high probability, the above algorithm returns a covering assignment.*

Proof. Consider an edge $uv \in E$. We have:

$$\begin{aligned}
\Pr[\lambda_A(u) \cap \lambda_A(v) = \emptyset] &= \prod_{i\in\lambda(u)\cap\lambda(v)} \Pr[t_i > 3\ln m \cdot \min\{x_{i,u}, x_{i,v}\}] \\
&= \prod_{i\in\lambda(u)\cap\lambda(v)} \{1 - 3\ln m \cdot \min\{x_{i,u}, x_{i,v}\}\} \\
&\leq \prod_{i\in\lambda(u)\cap\lambda(v)} \exp\{-3\ln m \cdot \min\{x_{i,u}, x_{i,v}\}\} \\
&= \exp\left\{-3\ln m \cdot \sum_{i\in\lambda(u)\cap\lambda(v)} \min\{x_{i,u}, x_{i,v}\}\right\} \\
&\leq \exp\{-3\ln m\} = \frac{1}{m^3}
\end{aligned}$$

By applying the union bound over all m edges we get that the algorithm returns a covering assignment with probability at least $1 - \frac{m}{m^3} = 1 - \frac{1}{m^2}$. ◀

Combining the above lemmas gives the claim. ◀

3.4 A k-approximation algorithm for the covering problem

Algorithm 2 A relaxation-based k-approximation algorithm.

1. Solve the LP relaxation and obtain a fractional solution $\{x_{i,v}\}_{i\in\lambda(v),v\in V}$.
2. Set $\lambda_A(v) = \{i \in \lambda(v) \colon x_{i,v} \geq \frac{1}{k}\}$, for every $v \in V$.
3. **return** $\{\lambda_A(v)\}_{v\in V}$.

▶ **Theorem 5.** *The above algorithm is a k-approximation algorithm for both the min-sum and the min-max covering assignment problems.*

Proof. Consider an edge $uv \in E$. By averaging, there exists an interface $i \in \lambda(u) \cap \lambda(v)$ such that $\min\{x_{i,u}, x_{i,v}\} \geq \frac{1}{k}$. Thus, $i \in \lambda_A(u) \cap \lambda_A(v)$ so λ_A is a covering assignment. Moreover, since $x_{i,v} \geq \frac{1}{k}$ for every $i \in \lambda_A(v)$, we have that $c(\lambda_A(v)) = \sum_{i\in\lambda_A(v)} c(i,v) \leq k \cdot \sum_{i\in\lambda_A(v)} x_{i,v} \cdot c(i,v)$. Thus, depending on the cost function, we have $\texttt{COST}(\lambda_A) \leq k \cdot \max_{v\in V} \sum_{i\in\lambda(v)} x_{i,v} \cdot c(i,v) = k \cdot \texttt{OPT}$. ◀

3.5 ILP formulation for the connection problem

In this section we again assume that $\texttt{OPT} \geq 1/2$, the cost of activating any interface is at most 1 and we distinguish between cheap and expensive vertices as before. We use a similar LP to model the connection problem, however we use additional flow variables to encode the connectivity requirement. The $x_{i,v}$ variables are defined as before. For each edge e we add an additional flow variable y_e which is 1 if and only if the edge e is covered by the solution. Connectivity of the terminals U is enforced via cut constraints: we require that for every cut $(S, V \setminus S)$ that separates at least one pair of terminals, the total y-weight of edges crossing the cut is at least 1. Then we can formulate the min-max connection problem as follows:

$$
\begin{aligned}
\min \quad & M \\
\text{s.t.} \quad & \sum_{i\in\lambda(v)} x_{i,v} \cdot c(i,v) \leq M \quad \forall v \in V \\
& \sum_{i\in\lambda(u)\cap\lambda(v)} \min\{x_{i,u}, x_{i,v}\} \geq y_{uv} \quad \forall uv \in E \\
& \begin{cases} x_{i,v} = 1 \quad \forall i \in \lambda(v) & \text{if } \sum_{i\in\lambda(v)} c(i,v) \cdot x_{i,v} < 1 \\ \sum_{i\in\lambda(v)} c(i,v) \cdot x_{i,v} \geq 1 & \text{otherwise} \end{cases} \quad \forall v \in V \\
& y(\delta(S)) \geq 1 \quad \forall S \subset V \colon S \cap U \neq \emptyset,\ (V \setminus S) \cap U \neq \emptyset \\
& x_{i,v} \in \{0,1\} \quad \forall v \in V, i \in \lambda(v) \\
& y_{uv} \in \{0,1\} \quad \forall uv \in E
\end{aligned}
$$

Observe that the above ILP has exponential number of constraints, but we can solve its LP relaxation in polynomial time by using the ellipsoid method and a separation oracle for the cut constraints, which can be implemented using a max-flow algorithm.

3.6 Randomized $O\left(\log^2 m\right)$-approximation algorithm for the connection problem

Algorithm 3 A relaxation-based $O(\log^2 m)$-approximation algorithm.

1. Solve the LP relaxation and obtain a fractional solution $\{x_{i,v}\}_{i\in\lambda(v),v\in V}, \{y_{uv}\}_{uv\in E}$.
2. For every edge $e \in E$, let $p_e = \min\{1, 3\ln m \cdot y_e\}$. Pick independent uniformly random variables Y_e such that $Y_e = 1$ with probability p_e and $Y_e = 0$ otherwise. Let H be the subgraph of G induced by the edges e such that $Y_e = 1$.
3. Use the rounding algorithm for the covering problem described on the LP solution restricted to the subgraph H to obtain a covering assignment λ_A for H.
4. **return** $\{\lambda_A(v)\}_{v\in V}$.

▶ **Theorem 6.** *There exists an $O\left(\log^2 m\right)$-approximation algorithm for both the min-sum and the min-max connection assignment problems which succeeds with high probability.*

Proof. We show that the algorithm for the covering problem described in the previous section can be used to obtain an $O\left(\log^2 m\right)$-approximation algorithm for the connection problem. We start with the following lemma:

▶ **Lemma 7.** *With high probability, the subgraph H connects all terminals in U.*

Proof. Fix any $S \subset V$ with $S \cap U \neq \emptyset$ and $(V \setminus S) \cap U \neq \emptyset$. Let $\ell \in \mathbb{N}$, such that $\ell \leq y(\delta(S)) < \ell + 1$. We have:

$$\begin{aligned}
\Pr[|\delta_H(S)| = 0] &= \prod_{e\in\delta(S)} \Pr[Y_e = 0] = \prod_{e\in\delta(S)} (1 - p_e) \\
&\leq \prod_{e\in\delta(S)} \exp\{-p_e\} \leq \exp\left\{-\sum_{e\in\delta(S)} \min\{1,\, 3\ln m \cdot y_e\}\right\} \\
&\leq \exp\{-3\ell \ln m\} = \frac{1}{m^{3\ell}}
\end{aligned}$$

where the last inequality uses the LP constraint $y(\delta(S)) \geq 1$ for every U-separating cut, so $\sum_{e\in\delta(S)} y_e \geq \ell$.

A cut in G is called an α-minimum cut if its value is less than α times the value of a minimum cut in G. We now make a use of the following Karger's lemma [13, 12]:

▶ **Lemma 8** (Karger's lemma). *The number of α-minimum cuts in a graph is at most $n^{2\alpha}$.*

Observe that by construction of the LP, every U-separating fractional cut has a value of at least 1, so there are at most $n^{2\ell}$ U-separating cuts of value less than ℓ. By applying the union bound over all U-separating cuts we get that:

$$\begin{aligned}
\Pr[\exists\, U\text{-sep. cut } S\colon |\delta_H(S)| = 0] &\leq \sum_{\ell=1}^{\infty} \sum_{\substack{S\subset V,\, S\cap U\neq\emptyset \\ (V\setminus S)\cap U\neq\emptyset,\; y(\delta(S))<\ell+1}} \Pr[|\delta_H(S)| = 0] \\
&\leq \sum_{\ell=1}^{\infty} n^{2\ell} \cdot \frac{1}{m^{3\ell}} = \sum_{\ell=1}^{\infty} \frac{1}{m^{\ell}} = \frac{1}{m-1}
\end{aligned}$$

Thus, with high probability, no U-separating cut is empty in H, i.e., H spans U. ◀

Now we want to show that by covering H we can obtain good approximation for connecting G. We have the following lemma:

▶ **Lemma 9.** *With high probability, the cost of covering H is at most $9 \ln m \cdot \mathtt{OPT}$.*

Proof. Let $\bar{\mathbf{x}}, \bar{\mathbf{y}}$ be the fractional solution such that every variable is scaled up by a factor of $3 \ln m$. Clearly, this is a valid LP solution. Moreover, by the linearity of the LP objective function, we know that the cost of $\bar{\mathbf{x}}, \bar{\mathbf{y}}$ is at most $3 \ln m \cdot \mathtt{OPT}$. Since for every edge $e \in E$, we have that $\Pr[Y_e = 1] = \min\{1, 3 \ln m \cdot y_e\}$, by linearity of expectations we know that the expected cost of covering H, call it $\mathtt{COST}(H)$ is at most $3 \ln m \cdot \mathtt{OPT}$. By applying the Chernoff bound with $\delta = 2$ we have that:

$$\Pr[\mathtt{COST}(H) \geq 3 \cdot \mathbb{E}[\mathtt{COST}(H)]] \leq \left(\frac{e^2}{3^3}\right)^{\mathbb{E}[\mathtt{COST}(H)]} \leq \exp\{-\mathbb{E}[\mathtt{COST}(H)]\} \leq \frac{1}{m^3}.$$

where in the last inequality we used the fact that due to our treatment of cheap and expensive vertices we get that $\mathbb{E}[\mathtt{COST}(H)] \geq 1$. Thus, with high probability, the cost of covering H is at most $9 \ln m \cdot \mathtt{OPT}$. ◀

Observe that the any LP solution restricted to a subgraph of G has cost at most the cost of the original solution. Since we know that the rounding algorithm for the covering problem returns a covering assignment of cost at most $9 \ln m \cdot \mathtt{OPT}_{LP}$ with high probability, where $\mathtt{OPT}_{LP}$ is the cost of the LP solution, we get that with high probability the cost of the returned connecting assignment is at most $O\big(\log^2 m\big) \cdot \mathtt{OPT}_{LP} \leq O\big(\log^2 m\big) \cdot \mathtt{OPT}$. ◀

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